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CIViC: Evidence Gates for Audio-Conditioned Visual Prediction on Proxy Tasks

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Abstract

Removing audio can shift a visual predictor's output without showing whether the task uses the intended audio information. We introduce Conditional Input and Visual Interpretation Checks (CIViC), a four-gate protocol that links an interpretation to an intervention, held-out task comparison, or endpoint control. We demonstrate CIViC on actor-disjoint face-raster tests, a synthetic audio-feature mixture, and source-video-disjoint SpeakerVid proxy tasks. In the static face-raster test, cross-actor audio replacement meets the registered detection rule with a small effect; on a separate temporal endpoint, constant-audio fits score better than audio-trained fits in four matched seeds. In the synthetic mixture, the condition-augmented fit has higher 1,000-step RMSE than the audio-only fit (0.1807 vs. 0.1707), with the gap shrinking during training. In the SpeakerVid proxy, the five-feature-zero fit (five low-level audio summaries zeroed; identity, dialogue, and condition inputs retained) has lower whole-frame grayscale RMSE than aligned audio across four held-out conditions and five paired seeds, while aligned audio has lower segment-drift error. The results show that output activation, held-out error, and endpoint controls support different interpretations across these proxy tasks.

1 Introduction

Audio-conditioned video and avatar models often expose paths for speech, speaker identity, timing, and structured scene information (Corona et al., 2025; Kong et al., 2025; Zhong et al., 2025; Peng et al., 2025; Chu et al., 2026). A common intervention removes or perturbs one path and checks whether the output changes. That test is useful, but it leaves several interpretations open. A model can react to audio removal while ignoring the identity of replacement audio. An auxiliary condition can alter predictions while worsening the held-out task. A proxy ordering can appear favorable even when a condition-only comparator predicts its target better than the audio-informed model.

This paper asks a narrow question: which checks distinguish input activation from a supported audio-conditioned visual prediction statement in these controlled tasks? CIViC answers this question with four gates. Each gate targets one alternative explanation. We fix the split, intervention, score, aggregation rule, and decision rule before held-out scoring. CIViC is not a video generator. It is a diagnostic framework that makes a conditional interpretation falsifiable before it is used to motivate a broader audiovisual statement. It does not establish that these four checks are necessary or sufficient outside the tested settings.

Prior audio-driven avatar systems evaluate generation, synchronization, or multi-person control (Corona et al., 2025; Kong et al., 2025; Zhong et al., 2025; Peng et al., 2025; Chu et al., 2026); other work studies disentangled audio representations or disrupts audio control through portrait perturbations (Mittal & Wang, 2020; Gan et al., 2025). MultiTalk targets audio–person binding, and DualTalk studies speaker–listener role transitions. CIViC does not rank those generators or assess their perceptual outputs. Its contribution is a mapping from a specified conditional interpretation to the matched intervention, aggregation unit, and endpoint needed to evaluate it; the case studies introduce no new architecture.

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The closest intervention protocol is the lip-leakage evaluation of Yaman et al. (2026), which tests silent input, mismatched audio, matched audio, and reference-frame selection on talking-face generators using LRS2. These tests already show why matched synthesis alone cannot identify audio control; our removal and replacement actions are not new. CIViC adds a held-out task comparison and a condition-only endpoint comparator within a registered interpretation rule. It does not evaluate the generators or perceptual endpoints in that work. Weng et al. separate speech, effects, and music into temporal controls (Weng et al., 2025); our feature-space study instead tests a withheld combination of synthetic perturbations and a factor vector against a compact proxy. Across these papers, the tasks and evidence differ; CIViC is a bounded interpretation-to-control mapping, not a new identification method or a general evaluation standard.

Table 1 makes the distinction from familiar evaluation actions explicit. Audio removal, replacement, held-out scoring, and shortcut controls each answer a useful but narrower question. CIViC does not replace any of them; it registers the interpretation for which each is needed and blocks that interpretation when its required evidence fails.

Table 1: From common evaluation actions to bounded interpretations. The entries are complementary, not a ranking of prior work.

Evaluation action	**What it can establish**	**What CIViC adds before the stated interpretation**
Audio removal or ablation	The registered path affects this output	A matched replacement for a value-specific statement
Input replacement	The stated alternative changes this endpoint	Held-out task evidence when benefit is asserted
Held-out task score	An ordering on the registered endpoint	A shortcut control for a transfer or proxy statement
Perturbation or defense test	Robustness to the tested disruption	A gate-specific stop rule and retained failed gate

For anonymous review, we provide the portable evaluation toolkit and lightweight checks through a 4open.science Git mirror.[1]

We apply CIViC to actor-disjoint face-raster checks, a synthetic feature-mixture task, and a source-video-disjoint SpeakerVid proxy study (Zhang et al., 2026). The studies retain a small static detection, an early proxy disadvantage that shrinks with fitting, and a source-video-disjoint endpoint warning. They do not establish general audio-conditioning benefit or failure. Disjointness is a split property: repeated seeds and hardware runs do not add independent actors or source videos.

Contributions. The paper makes four contributions:

1. CIViC, a four-question diagnostic framework that separates input activation, substitution specificity, held-out benefit, and endpoint validity;
2. actor-disjoint CREMA-D raster checks: the static protocol satisfies its registered detection rule but has a very small cross-actor audio effect, while a separate temporal renderer does not establish an audio benefit on its endpoint;
3. a synthetic audio-feature study over 205/24/27 source clips, with zero cross-split source-clip, original-video-group, video-namespaced-speaker, and exact-target overlap, showing an active condition branch, a shrinking error gap with training, policy-dependent controls, and a condition-only warning on a separate scalar endpoint; and
4. a new source-video-disjoint SpeakerVid proxy study with a five-arm, five-seed control matrix, four-platform execution repeats, and a post hoc check across its four held-out source-video groups.

Table 2 records the decision rule. A detection excludes only its named alternative; broader or practically important interpretations also need a task-scale threshold or equivalence region registered

[1]Anonymous repository and documentation: 4open.science project page; source code: download the code-only ZIP, also provided as a separate supplementary file. The archive includes deterministic fixtures and a pinned lightweight environment, but not restricted study inputs or assets needed to replay the full experiments.

with its source, unit, direction, and failure action. No threshold was registered for the tiny CREMA-D cross-actor contrast, so it remains practically unclassified. The three-seed Study II result is a fixed-run direction without a held-out-case interval or population inference; zero cross-split overlap establishes split provenance, not a population interval.

Table 2: Interpretation rules. A detection result excludes only the named alternative; practical use needs a task-scale rule.

Gate	Required evidence	What a detection does not establish
Activation	Output changes after the registered removal	Which input values the model uses
Substitution	Matched replacement changes the stated endpoint	Task benefit or general audio use
Benefit	Matched held-out error comparison	A mechanism for any observed gap
Endpoint validity	Control excludes the named shortcut	Validity outside the checked endpoint

2 CIViC maps interpretations to four evidence questions

Let $f(x, a, z)$ map a fixed visual context x, audio a, and optional auxiliary condition z to a motion target y. A nonzero difference between $f(x, a, z)$ and $f(x, 0, z)$ only shows that removing audio changes the output. Figure 1 maps four CIViC gates to the alternative explanation each gate excludes. The gates are not a universal checklist. A study runs the gates required by its intended interpretation. For example, a speaker-specific statement needs a registered speaker-audio substitution, a mixed-condition benefit needs a matched task comparison, and a proxy-transfer statement needs an endpoint control that excludes non-audio shortcuts.

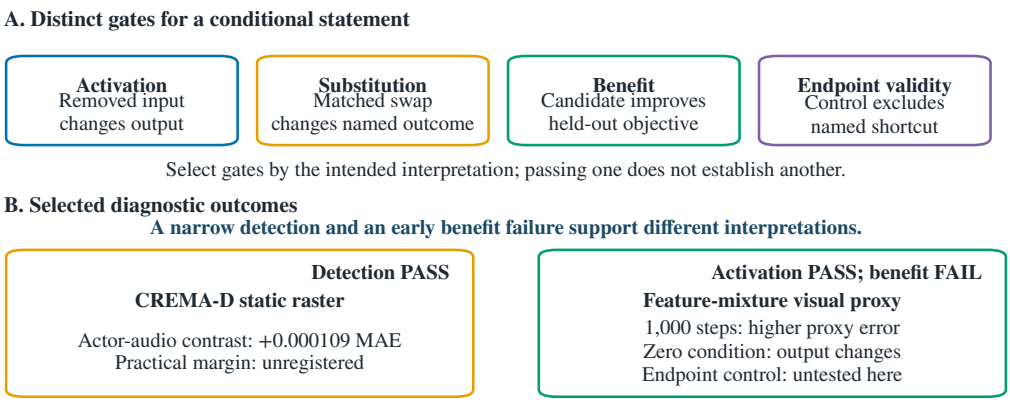

Figure 1: Four interpretation gates (A) and two selected diagnostic summaries (B). CREMA-D meets its registered fixed-pair detection rule but has no practical threshold. In the synthetic feature-mixture proxy, the condition changes predictions while early held-out error is higher; that gap shrinks with training. Its matched endpoint control is untested. The newer SpeakerVid proxy comparison is described in Section 5, not shown here. Color and text distinguish the gates.

The gates are selected for the intended interpretation and fix the split, intervention, score, aggregation unit, and decision rule before held-out scoring. An endpoint warning excludes the named interpretation but does not identify its shortcut; a new control must exclude the competing endpoint result before that interpretation is restored. Repeated fits measure run variation, not added population coverage. For a practically useful actor-matched raster statement, the protocol card therefore requires an audio replacement, a held-out score, a task-scale threshold, and an endpoint control when the target is a proxy. Figure 1 contrasts the narrow CREMA-D detection with the failed early-benefit check; Table 3 gives the paired raster results.

3 Study I: static and temporal face-raster checks

3.1 CREMA-D meets a narrow fixed-pair detection rule

The first raster application uses CREMA-D speech-video recordings (Cao et al., 2014). A fixed protocol assigns 73 actors to development, 9 to validation, and 9 to sealed confirmation. The renderer receives an identity raster decoded at 0.25 seconds and a waveform-derived log-mel window from 0.75 to 1.25 seconds. It predicts the 64×64 target raster at 1.0 seconds. The audio window extends 0.25 seconds beyond the target time; this is a centered-window raster reconstruction. Inputs exclude emotion, intensity, sentence, and demographic labels. For this raster application, mean absolute error (MAE) averages absolute RGB intensity differences over all $64 \times 64 \times 3$ values after division by 255, so it is measured on a $[0, 1]$ channel scale. We set the confirmation protocol, pairings, and bootstrap rule before evaluating the 732 confirmation clips. For actor-audio replacement, actors follow a fixed cycle, and a deterministic mapping from protocol and clip identity selects a clip from the next actor. Utterance, emotion, and intensity are not matched, so this control changes speaker and speech content together.

Table 3 reports the fixed paired contrasts. Every registered lower interval bound is positive. The cross-actor-audio effect is 0.000109 MAE, however. This meets the registered sign-based detection rule, but no task-scale threshold was registered, so its practical relevance is unclassified. It does not establish robust speaker transfer, temporal generation, natural multi-person behavior, active-speaker accuracy, or human-perceptual quality.

Table 3: CREMA-D static confirmation on 732 fixed pairs from an actor-disjoint split. Differences are control MAE minus aligned MAE; aligned MAE is 0.02907. Paired intervals use 10,000 clip-level resamples with seed 1701 for the fixed checkpoint and pair set; they do not quantify actor-population uncertainty. Positive differences favor aligned input.

Control	**Control MAE − aligned MAE ↑**	**Paired 95% interval**
Zero audio	+0.131475	[0.127163, 0.135640]
Actor-audio replacement	+0.000109	[0.000034, 0.000186]
Identity-raster replacement	+0.064860	[0.062795, 0.066897]

3.2 A separate temporal renderer fails its audio controls

The static test does not answer whether audio improves a sequence renderer. A separately registered renderer was fitted on 5,038 development sequences, selected on 689 validation sequences, and evaluated on 708 sealed sequences. Aligned MAE was 0.030449; shifting audio raised it by 0.0000176, while repeating the identity raster lowered it by 0.010592. Without an interval or practical margin for the shift, this check does not establish a sequence-level audio benefit or measure perceptual video quality.

3.3 A separate confirmation fails the audio-replacement gate

A second actor-disjoint raster protocol uses the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) (Livingstone & Russo, 2018). An earlier result summary scored 1.0 seconds rather than the registered 2.0-second target, so we evaluated the 240-pair protocol and checkpoint at 2.0 seconds. Aligned MAE is 0.08990. Zero audio raises it by 0.20931 (paired 95% interval $[0.20469, 0.21358]$), while another actor's audio changes it by -0.00045 ($[-0.00129, 0.00039]$); the registered replacement rule fails. Exploratory filename-matched cross-actor and same-actor statement swaps change MAE by -0.00102 ($[-0.00491, 0.00141]$, four actors) and $+0.00004$ ($[-0.00017, 0.00025]$), respectively.

A same-architecture constant-audio fit scores 0.08652 on those targets, versus 0.08990 for the audio-trained fit (difference -0.00338, descriptive four-actor bootstrap 95% interval $[-0.00512, -0.00164]$). Three matched-seed fits also favor constant audio; all four differences range from -0.00338 to -0.00216. A fresh four-seed, 80-epoch repeat favors constant audio in all four pairs, but trajectories diverge before epoch 40, so it does not isolate extra fitting. These retrospective comparisons do not create a registered benefit gate. Removal is activation-sensitive, but replacement fails and neither

swap isolates speaker identity because speech content also changes. Appendix B gives the registered endpoint and within-trajectory details.

3.4 An incompatible external audit tests a separate boundary

We apply the frozen renderer to 33 AVA Active Speaker validation recordings (Roth et al., 2020). The shared soundtrack has no per-face audio, so this is an external boundary check only. Aligned, shifted, and zero-audio AUROC are 0.63678, 0.63670, and 0.63672; the aligned-minus-shifted interval is $[-0.00017, 0.00028]$. This check supplies no speaker-attribution evidence.

4 Study II: an active condition offers no stable benefit on a withheld feature mixture

4.1 Controlled composition protocol

Study II uses 819 indexed synthetic feature mixtures from a private, provenance-approved source family: 615 train, 96 validation, and 108 test records expanded from 205/24/27 source clips. The train, validation, and test splits have zero pairwise overlap in source clips, original-video groups, video-namespaced speakers, and exact target face arrays. These are provenance and split checks, not independent-sample or population claims. Each mixture starts with five speech summaries and adds the same nuisance sequence to both speakers. Environment-like features are seeded normalized noise; music-like features are seeded sinusoids, with either slow/transient noise and an optional beat envelope. They are five-coordinate feature arrays, not waveforms, scaled to 0.25 and 0.20 times speech-feature standard deviation.

The source feature index lists 5-dimensional audio, 6-dimensional dialogue, 128-dimensional face, and 256-dimensional identity streams, all at 25 Hz. The 128-coordinate target is copied from the source speaker-axis pack. Its extractor decodes whole frames at 25 Hz with `ffmpeg_gray_proxy`, scales each frame to 16-by-16 grayscale, averages adjacent vertical pixel pairs to obtain an 8-by-16 array, and subtracts 0.5. Speaker-axis construction gates the face target with $0.20 + 0.80a$, where a is annotation-derived activity; MixText copies this target into each condition. Streams are paired by frame index and truncated to their common frame count, with no presentation-timestamp sidecar retained. The identity proxy is the mean of the first eight grayscale frames repeated over speaker slots. RMSE is computed over these dimensionless coordinates without per-condition rescaling; it is not a physical motion or video-quality measure.

Both arms receive five audio summaries, this image-derived 256-coordinate identity proxy, and six annotation-derived dialogue coordinates. The splits have zero overlap in source clips, original-video groups, and video-namespaced speakers, but real-person identity disjointness remains unverified. The added condition is the seven-bit vector $[s, e, m, e_0, e_1, m_0, m_1]$, encoding speech, environment, music, and two type indicators per nuisance stream. The shared past-only backbone uses a 96-frame context, width 256, and four attention heads. Adding the condition increases input width; the joint mixed condition is withheld from training. The primary comparison is the 1,000-step checkpoint, with the same three seeds checked at 3,000 and 10,000 steps. Separate analyses test training with all conditions present, 24-seed training-support policies, five-seed parameter-matched controls, and source-disjoint scalar endpoint shortcuts; their estimands are not pooled. The registered fixed-run rule requires positive structured-minus- audio gaps for all three seeds to indicate no benefit at 1,000 steps. This is a directional result without a held-out-case interval or population inference.

4.2 The early error gap shrinks with training

Table 5 reports three checkpoints from the same runs. At 1,000 steps, all three paired mixed-condition gaps are positive; by 10,000 steps the mean is 0.0019 ± 0.0032 and one seed gap is negative. The early disadvantage therefore shrinks with fitting. Figure 2 shows that zeroing the condition changes predictions by 0.2123 ± 0.0070 and replacing the mixed vector with the clean vector changes them by 0.1473 on average; neither displacement measures task benefit. Clean, environment-only, and music-only test gaps are -0.000284, 0.001008, and 0.000909, versus +0.010032 on the withheld mixture. Relative to clean, mixture error rises by +0.0095 for audio only and 0.0198 with the condition. The 24-seed support audit has a positive structured-minus-audio gap for every policy (Table 4). With all conditions in training, a separate three-seed control gives mixed RMSE 0.1659 ± 0.0027 for audio only and 0.1684 ± 0.0047 with the condition; it does not test withheld composition.

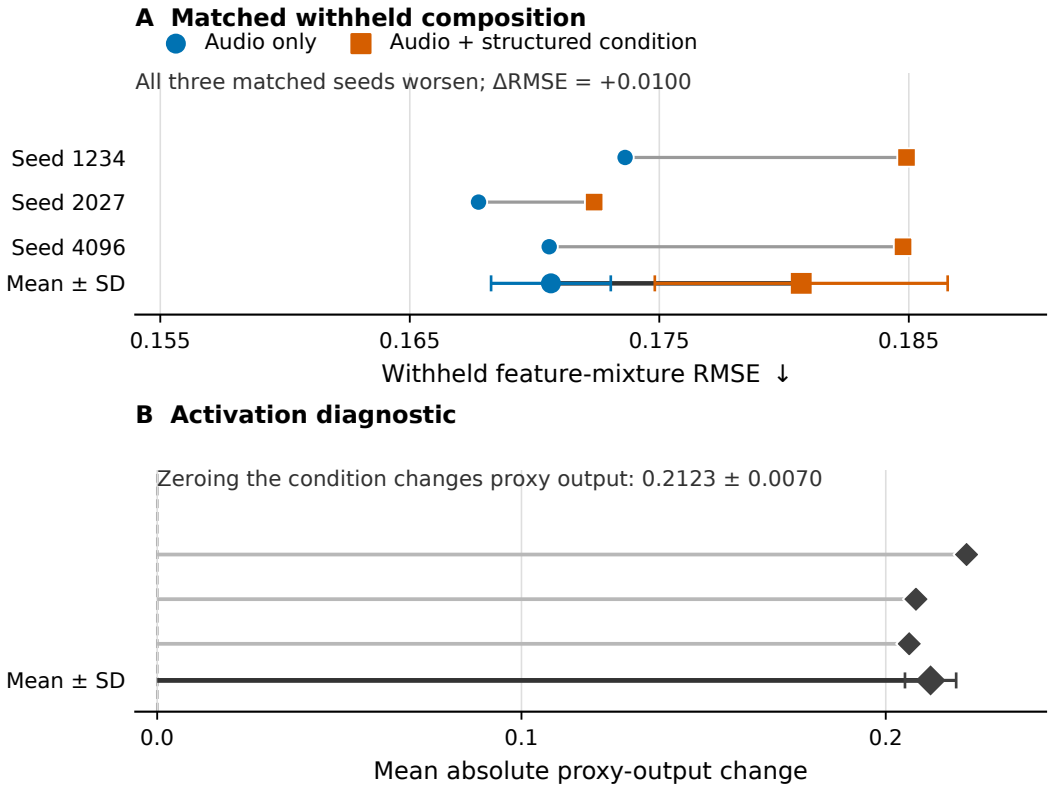

Figure 2: Registered 1,000-step feature-mixture evidence. (A) Adding the condition raises withheld-composition RMSE in all three matched seeds; the target is a dimensionless stored proxy. (B) Zeroing the condition changes proxy predictions in the same runs. Points are seed values; markers are mean ± standard deviation. Activation is not a task-error comparison.

Table 4: Twenty-four-seed, 1,000-step training-support audit on the held-out mixed condition (mean ± population standard deviation). The gap is added-condition minus audio-only RMSE; positive gaps favor audio only. Each policy changes training support and is not pooled with the three-seed result.

Training support	**Audio-only RMSE ↓**	**Added-condition RMSE ↓**	**Gap ↓**
Single clean	0.4249 ± 0.0097	0.6921 ± 0.0218	+0.2672 ± 0.0276
Single environment	0.1638 ± 0.0026	0.4192 ± 0.0239	+0.2554 ± 0.0240
Single music	0.1681 ± 0.0025	0.4169 ± 0.0247	+0.2487 ± 0.0252
Clean plus environment	0.1685 ± 0.0027	0.4121 ± 0.0225	+0.2436 ± 0.0231

Table 5: Withheld mixed-composition error at three checkpoints of the same three-seed runs (mean ± population standard deviation; lower RMSE is better). Audio only omits the seven-bit condition; both arms retain speaker and dialogue inputs. The gap is added-condition minus audio-only RMSE; positive favors audio only.

Steps	**Audio-only RMSE ↓**	**Added-condition RMSE ↓**	**Gap ↓**
1,000	0.1707 ± 0.0024	0.1807 ± 0.0059	+0.0100 ± 0.0040
3,000	0.1338 ± 0.0004	0.1366 ± 0.0023	+0.0028 ± 0.0019
10,000	0.1237 ± 0.0014	0.1256 ± 0.0042	+0.0019 ± 0.0032

4.3 A parameter-matched audit exposes a training-support mismatch

The primary comparison changes input width, so we separately evaluated three parameter-matched control policies across five seeds under clean-plus-environment support. Each uses the seven-coordinate interface and 4,908,416 parameters (Table 6). The dummy-condition model scores 0.1822 ± 0.0025; training with shuffled conditions but aligned evaluation scores 0.4248 ± 0.0217; and five-feature-zero with aligned conditions scores 0.5003 ± 0.0278. These policies change multiple inputs or train–evaluation support, so their gaps isolate neither audio nor condition alignment. The same-checkpoint intervention in Figure 2 supplies the condition-activity test. These controls neither make the primary three-seed comparison parameter matched nor show benefit. We exclude a train-by-evaluation comparison because its underlying results could not be verified.

Table 6: Five-seed parameter-matched controls on the held-out mixture under clean plus environment training support (mean ± population standard deviation; seeds 1234, 2027, 4096, 8191, 16384). All entries use seven condition coordinates and 4,908,416 parameters. Shuffled values are used in training but aligned values in evaluation; this is a policy-specific diagnostic.

Control	**Audio**	**Condition relation**	**Mixed RMSE ↓**
Dummy condition	Observed	Deterministic dummy	0.1822 ± 0.0025
Shuffled condition	Observed	Train shuffled; test aligned	0.4248 ± 0.0217
Five-feature-zeroed	Zeroed	Aligned	0.5003 ± 0.0278

4.4 A condition-only comparator raises an endpoint warning

A source-file-disjoint CREMA-D ridge check (71 clips, three held-out actors, five mixtures) gives RMSE 0.00190136 for the interaction model versus 0.00189705 for audio only; no paired interval or practical margin was retained. A separate RAVDESS ridge check over 1,441 clips and 24 source groups gives structured-minus-audio RMSE -0.00000924 with interval $[-0.00002350, 0.00000148]$; metadata-only RMSE (0.00881499) is below both audio-informed variants (0.00926477 and 0.00927402). Neither check isolates audio use in the scalar proxy.

A separate source-disjoint neural follow-up fits fixed width-32 ReLU regressors on CREMA-D and evaluates four held-out GRID talkers. Its structured-minus-audio contrast is -0.0023303 with talker-clustered interval $[-0.0032104, -0.0006693]$, but a condition-only comparator reaches RMSE near 0.00109, below the structured (0.00865–0.01341) and audio-only (0.01231–0.01592) models. These descriptive scalar-proxy checks show that an apparently favorable ordering cannot support the intended audio-conditioning interpretation; they do not identify the shortcut in Study II's feature-mixture comparison.

5 Study III: a source-video-disjoint SpeakerVid proxy study

Study II's primary target also has 128 coordinates, but its separate shortcut follow-ups use a scalar proxy. We run a new source-video-disjoint study on SpeakerVid-5M (Zhang et al., 2026). The study uses 865 decoded two-speaker windows from 31 videos (649/108/108 train/validation/test records across 23/4/4 videos). Source-clip, source-video, and exact-target overlap are zero between splits. One unverified video was excluded; this is a new source pack, not a replay of Study II. It supplies no matched comparator for that study.

Each record has five low-level audio summaries, an image-derived identity descriptor, annotation-derived dialogue features, and a 128-coordinate visual target. The five audio summaries are log energy, root-mean-square amplitude, zero-crossing rate, mean, and standard deviation, computed on 40-ms windows with 40-ms hops (25 features per second). The video extractor samples whole frames at a nominal 25 Hz, scales each to 16-by-16 grayscale, averages adjacent vertical pixel pairs, and subtracts 0.5 to form an 8-by-16 (128-value) raster. It uses the whole frame, not a face or mouth crop, landmarks, or a semantic motion representation. The feature files retain the nominal frame rate but no presentation timestamps, so streams are paired by frame index and truncated to their common frame count. The six dialogue coordinates are two speaker-activity indicators, active-speaker count, turn-start, turn-end, and overlap, derived from supplied annotations. The 256-value identity input is the mean 16-by-16 grayscale frame over the first eight frames, repeated over time and speaker slots; it is not a pretrained

speaker embedding. In the speaker-axis construction, audio is activity-weighted and the visual target is scaled by a gate of $0.20 + 0.80a$, where a is annotation-derived speaker activity. The target and remaining dialogue input therefore share activity information. This endpoint does not establish mouth motion or audio-video synchrony. A deterministic feature-mixture generator provides clean, environment-like, and music-like training conditions plus a held-out mixture; its seven-dimensional vector encodes synthetic presence/type, not a waveform or natural-language prompt. Five trained arms are aligned audio summaries and condition, five-feature-zero with aligned condition, observed summaries with the condition branch omitted, observed summaries with a dummy condition, and observed summaries with batch-shuffled conditions. They share the data, targets, split, 1,000-step schedule, and five seeds (1234, 2027, 4096, 8191, 16384).

The five-feature-zero arm sets the five audio-summary inputs to zero while retaining the image-derived identity descriptor, annotation-derived dialogue features, and aligned condition vector. Because that condition vector encodes synthetic speech presence and type, the arm is not a silent-input condition and does not remove all audio-derived information. We repeated the full matrix on four accelerator platforms: one H100, two A100 (labeled A100-1 and A100-2 in Table 7), and one V100. The 100 fits comprise 25 fits per platform and reuse one source pack across four platforms; they are compute replications, not new independent evidence. A post-hoc evaluation groups aligned-audio and five-feature-zero predictions by the four held-out videos; it describes those groups only.

5.1 Five-feature-zero has lower test error across platforms and source groups

Table 7 reports the paired test difference in root mean squared error (RMSE) over the 128 grayscale-raster coordinates, computed as control minus aligned audio for each seed. Negative values favor the control. Five-feature-zero has a negative mean in all 16 platform-by-condition cells, and all five paired seeds are negative in each cell. On validation, the same contrast is negative for all five seeds in every condition on every platform.

This ranking is specific to raster-reconstruction RMSE. A secondary segment-drift proxy, for which lower is better, has the opposite mean ordering for five-feature-zero minus aligned audio in all 16 test platform-condition cells. The four-platform ranges of five-seed means are $[0.0057, 0.0064]$ for clean, $[0.0076, 0.0079]$ for environment-like, $[0.0069, 0.0070]$ for music-like, and $[0.0043, 0.0050]$ for mixed conditions. These are repeated fits on one source pack, not intervals over new videos. The evaluator flattens the predicted feature windows, divides that array into 10-second segments, and compares later segment means with the first; it does not track drift in a continuous rendered clip. Appendix A, Table 8, reports the contrasting proxy alongside raster-reconstruction RMSE and its seed-level signs. Neither proxy establishes rendered-video quality; the drift result prevents an unqualified statement that five-feature-zero is better.

Table 7: Paired test raster-reconstruction RMSE differences for the five-feature-zero control, computed as five-feature-zero minus aligned audio. Entries are mean ± sample standard deviation over five paired seed differences. Negative values favor five-feature-zero. The four platform runs share one source pack and are hardware or execution replications, not independent data. The secondary segment-drift proxy favors aligned audio in every condition; Table 8 reports that opposing result.

Condition	**H100 ΔRMSE↓**	**A100-1 ΔRMSE↓**	**A100-2 ΔRMSE↓**	**V100 ΔRMSE↓**
Clean	−0.0142 ± 0.0036	−0.0143 ± 0.0035	−0.0143 ± 0.0035	−0.0142 ± 0.0036
Environment-like	−0.0214 ± 0.0022	−0.0215 ± 0.0022	−0.0215 ± 0.0022	−0.0216 ± 0.0024
Music-like	−0.0210 ± 0.0019	−0.0211 ± 0.0019	−0.0211 ± 0.0019	−0.0213 ± 0.0020
Mixed	−0.0206 ± 0.0024	−0.0208 ± 0.0024	−0.0208 ± 0.0024	−0.0212 ± 0.0027

The audio-only model, which keeps observed speech but omits the condition branch, has lower mean test RMSE in all 16 platform-condition cells. All five seed differences are negative except music, where three platforms have four negative seeds and V100 has five. Because input dimensionality changes, this is a branch-omission comparison, not an isolated audio effect. The dummy-condition model has higher RMSE in every cell and seed. Shuffled conditions have small positive test means (0.0006–0.0022) but variable seed signs. On validation, audio-only means are lower in all cells (four of five negative seeds for clean, all five for the other conditions); dummy-condition means are higher in every cell and seed. Shuffled-condition means are positive in all 16 validation cells,

but seed signs vary. Validation remains separate from the test result. Appendix A, Table 9, shows the full test-condition contrast: the audio-only arm favors omission of the condition branch. The dummy-condition arm has higher raster-reconstruction RMSE, while the shuffled-condition arm has small positive mean differences and variable seed signs. Their secondary segment-drift means have a different ordering, detailed in Appendix A; this is not a single-metric ranking of the complete system.

The post-hoc grouping has 108 unique test windows from four videos, each scored in four conditions (432 video-condition rows). Every video-condition mean and all five seed differences are negative on each platform; means range from -0.0334 to -0.0123. These four groups are the complete held-out video subset, so this is descriptive consistency, not a population interval. Appendix A, Table 10, separates the four source videos; no one video reverses the five-feature-zero ordering in any condition, but these are not four independent test populations.

On test, replacing the aligned condition vector with zero moves predictions by 0.2127–0.2129 (seed SD 0.0115–0.0117) in aligned fits and 0.2614–0.2619 (SD 0.0104–0.0105) in five-feature-zero fits. Replacing it with the clean vector moves them by 0.1570 (SD 0.0105) and 0.1338–0.1341 (SD 0.0080), respectively. These are displacements, not condition correctness or task benefit; text fields are absent for audio-only because it has no condition branch.

Raster-reconstruction RMSE is the primary endpoint. Active/inactive RMSE are components; causal selectivity, target fidelity, and listener responsiveness are derived; non-target leakage duplicates inactive RMSE; and speaker-order delta is a raw perturbation diagnostic. Identity stability measures input-order invariance, not identity fidelity, while the coarse CIViC score is uncalibrated. For five-feature-zero fits, audio-video sync and modality consistency use a zero-energy fallback of 0.5, and lag is at its -400 ms search boundary; none is interpreted as synchronization performance. Turn-timing error remains a secondary proxy output. The bounded result is an endpoint-specific ordering for this compact target: the separately trained five-feature-zero fit has lower raster-reconstruction RMSE, while aligned audio has lower segment-drift proxy error on these held-out records. This comparison establishes no natural-video quality or validity for Study II's separately sourced target.

The Study III target contains 299,212 training and 50,788 test frame–speaker observations from 23 and four videos, respectively. The median of the 128 coordinate-wise population standard deviations was 0.2047 (train) and 0.3012 (test), computed over clean observations with the observation count as divisor. Test adjacent-frame RMS was 0.0198, the square root of mean squared differences between consecutive frame indices within records, pooled over both speaker slots and all coordinates. A coordinate-wise mean fitted on clean training frames scored RMSE 0.3477 over 424 clean test windows using the model's 96-frame window, 48-frame stride, final-window rule, and coordinate-wise aggregation (10,420,224 values). Five-seed clean-test mean RMSE was 0.2005 for aligned audio and 0.1864 for five-feature-zero. Both fitted predictors improve on this constant-mean baseline for the raster endpoint; this does not isolate audio, since five-feature-zero retains identity, dialogue, and aligned-condition inputs. The pooled target mean also shifts from -0.1526 in training to 0.0114 in test, so this baseline comparison is descriptive under a changed target distribution, not a measure of audio-specific predictability. These dimensionless statistics show raster variation, not mouth motion or audio-predictable dynamics: without source media or presentation timestamps, frame-index differences establish neither temporal motion nor audio alignment. Video-namespaced speaker labels do not overlap across splits, but real-person disjointness is unverified. Study II has zero source-clip and original-video- group overlap and uses a copied grayscale target; neither split check verifies real-person identity or natural-video validity.

6 Conclusion

These studies show that output activation, held-out error, and endpoint controls support different interpretations. CREMA-D's cross-actor effect is small and utterance-confounded under a detection rule with no practical threshold. In Study II, the early fit gap narrows with training, and same-checkpoint condition replacements displace predictions without measuring task benefit. Separate scalar proxy controls produce endpoint warnings that do not identify a shortcut in the feature-mixture target. In Study III, the separately trained five-feature-zero arm has lower raster RMSE than the aligned-audio arm, while aligned audio has lower segment-drift error. The audio-only branch-omission fit also has lower raster RMSE, but changes input width and therefore does not isolate a condition effect. The whole-frame target and four-video test set support descriptive proxy findings, not rendered-video

quality, natural-mixture performance, or speaker attribution. Human perceptual evaluation remains future work.

Reproducibility Statement

Sections 2–5 specify the protocols and bounded results. The anonymous code supplement provides deterministic checks, media-free fixtures, and a pinned lightweight environment; full experiment replay requires approved source material that the archive does not include.

AI Use Statement

OpenAI Codex assisted with literature search, manuscript review and editing, and preparation of the final source and PDF packages. Gemini and Claude reviewed wording in earlier drafts. The authors take responsibility for the final text and citations.

References

AMI Project. The AMI meeting corpus. University of Edinburgh corpus site, 2005. URL https://groups.inf.ed.ac.uk/ami/corpus/.

Houwei Cao, David G. Cooper, Michael K. Keutmann, Ruben C. Gur, Ani Nenkova, and Ragini Verma. Crema-d: Crowd-sourced emotional multimodal actors dataset. *IEEE Transactions on Affective Computing*, 5(4): 377–390, 2014.

Xuangeng Chu et al. Unils: End-to-end audio-driven avatars for unified listening and speaking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Chu_UniLS_End-to-End_Audio-Driven_Avatars_for_Unified_Listening_and_Speaking_CVPR_2026_paper.html.

Martin Cooke, Jon Barker, Stuart Cunningham, and Xu Shao. An audio-visual corpus for speech perception and automatic speech recognition. *The Journal of the Acoustical Society of America*, 120(5):2421–2424, 2006. doi: 10.1121/1.2229005.

Enric Corona et al. Vlogger: Multimodal diffusion for embodied avatar synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Corona_VLOGGER_Multimodal_Diffusion_for_Embodied_Avatar_Synthesis_CVPR_2025_paper.html.

Yuan Gan, Jiaxu Miao, Yunze Wang, and Yi Yang. Silence is golden: Leveraging adversarial examples to nullify audio control in ldm-based talking-head generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13434–13444, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Gan_Silence_is_Golden_Leveraging_Adversarial_Examples_to_Nullify_Audio_Control_CVPR_2025_paper.html.

Zhe Kong et al. Let them talk: Audio-driven multi-person conversational video generation. In *Advances in Neural Information Processing Systems*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/66fdfe692a360dfbad1fba32f64694da-Abstract-Conference.html.

Steven R. Livingstone and Frank A. Russo. The ryerson audio-visual database of emotional speech and song (ravdess): A dynamic, multimodal set of facial and vocal expressions in north american english. *PLOS ONE*, 13(5):e0196391, 2018.

Gaurav Mittal and Baoyuan Wang. Animating face using disentangled audio representations. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 3290–3298, 2020. URL https://openaccess.thecvf.com/content_WACV_2020/html/Mittal_Animating_Face_using_Disentangled_Audio_Representations_WACV_2020_paper.html.

Ziqiao Peng et al. Dualtalk: Dual-speaker interaction for 3d talking head conversations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Peng_DualTalk_Dual-Speaker_Interaction_for_3D_Talking_Head_Conversations_CVPR_2025_paper.html.

Joseph Roth et al. Ava-activespeaker: An audio-visual dataset for active speaker detection. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2020. URL https://research.google/pubs/ava-activespeaker-an-audio-visual-dataset-for-active-speaker-detection/.

Shuchen Weng, Haojie Zheng, Zheng Chang, Si Li, Boxin Shi, and Xinlong Wang. Audio-sync video generation with multi-stream temporal control. In *Advances in Neural Information Processing Systems*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/133239b0506b84c802a12b0e5a764a17-Abstract-Conference.html.

Dogucan Yaman, Fevziye Irem Eyiokur, Hazim Kemal Ekenel, and Alexander Waibel. Assessing identity leakage in talking face generation: Metrics and evaluation framework. *arXiv preprint arXiv:2511.08613*, 2026. URL https://arxiv.org/abs/2511.08613. Accepted to ICASSP 2026.

Youliang Zhang, Zhaoyang Li, Duomin Wang, Jiahe Zhang, Deyu Zhou, Zixin Yin, Xili Dai, Gang Yu, and Xiu Li. Speakervid-5m: A large-scale high-quality dataset for audio-visual dyadic interactive human generation. In *International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=U004uqALWl.

Zhimeng Zhang, Lincheng Li, Yu Ding, and Changjie Fan. Flow-guided one-shot talking face generation with a high-resolution audio-visual dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3661–3670, 2021.

Zhizhou Zhong, Yicheng Ji, Zhe Kong, Yiying Liu, Jiarui Wang, Jiasun Feng, Lupeng Liu, Xiangyi Wang, Yanjia Li, Yuqing She, Ying Qin, Huan Li, Shuiyang Mao, Wei Liu, and Wenhan Luo. Anytalker: Scaling multi-person talking video generation with interactivity refinement. *arXiv preprint arXiv:2511.23475*, 2025. URL https://arxiv.org/abs/2511.23475.

A SpeakerVid test contrasts by condition and source video

The test endpoint here is root mean squared error over the dimensionless 128-coordinate whole-frame grayscale proxy. Each difference subtracts the aligned-audio arm's error from the control arm's error on the same held-out records and paired training seed. Negative differences favor the control. The four accelerator runs reuse one source pack and the same five seeds; their spread is descriptive hardware sensitivity, not a confidence interval over videos.

Table 8 separates the two endpoints. Five-feature-zero has lower raster-reconstruction error, while aligned audio has lower error on the flattened-window segment-drift proxy in all four conditions. Positive paired drift differences occur in four of five seeds for clean and mixed conditions, and all five seeds for environment-like and music-like conditions. These ranges summarize four platforms on the same videos and are not independent-video intervals; flattened windows do not measure continuous-video quality.

Table 8: Five-feature-zero minus aligned-audio test differences for two lower-is-better proxies. Negative raster-reconstruction RMSE differences favor five-feature-zero, while positive drift differences favor aligned audio. The flattened-window drift statistic does not measure continuous-video quality. Ranges span four platform means computed from the same five seeds and four-video test subset; they are descriptive, not confidence intervals.

Condition	**Raster ΔRMSE ↓**	**Drift Δ ↓**	**Positive drift seeds**
Clean	$[-0.0143, -0.0142]$	$[0.0057, 0.0064]$	4/5
Environment-like	$[-0.0216, -0.0214]$	$[0.0076, 0.0079]$	5/5
Music-like	$[-0.0213, -0.0210]$	$[0.0069, 0.0070]$	5/5
Mixed	$[-0.0212, -0.0206]$	$[0.0043, 0.0050]$	4/5

Table 9 separates four controls that change different things. Zeroing the five audio summaries during fitting has lower error in every condition, and removing the condition branch also lowers the mean error. The dummy condition raises error for every seed. Shuffling condition vectors raises the mean error slightly, but the seed-level direction varies. These comparisons cannot be collapsed into one estimate of audio or condition use: the controls differ in fitted inputs, branch capacity, or condition pairing. The secondary segment-drift proxy adds a further qualification. Relative to aligned audio, audio-only lowers both mean raster-reconstruction RMSE and mean drift in every platform-condition cell. Dummy and shuffled conditions raise mean raster-reconstruction RMSE but lower mean drift in every such cell. For example, across the four platforms in the mixed condition, dummy-minus-aligned drift ranges from -0.0017 to -0.0015 and shuffled-minus-aligned drift from -0.0044 to -0.0041. These are descriptive five-seed means on the same four-video test subset, not independent-video intervals. Drift is computed over flattened feature windows, so the opposing directions do not establish improved continuous-video quality.

Table 9: Test raster-reconstruction RMSE difference (control minus aligned audio) by condition. Each cell is the minimum–maximum of four platform-level means, each computed from five paired seed differences. These ranges are not confidence intervals; platforms share the same held-out videos and seeds. Negative values favor the named control.

Condition	**Five-feature-zero ΔRMSE ↓**	**Audio-only ΔRMSE ↓**	**Dummy ΔRMSE ↓**	**Shuffled ΔRMSE ↓**
Clean	$[-0.0143, -0.0142]$	$[-0.0040, -0.0037]$	$[0.0165, 0.0167]$	$[0.0006, 0.0007]$
Environment-like	$[-0.0216, -0.0214]$	$[-0.0045, -0.0042]$	$[0.0165, 0.0167]$	$[0.0011, 0.0012]$
Music-like	$[-0.0213, -0.0210]$	$[-0.0046, -0.0042]$	$[0.0163, 0.0166]$	$[0.0011, 0.0012]$
Mixed	$[-0.0212, -0.0206]$	$[-0.0106, -0.0101]$	$[0.0108, 0.0111]$	$[0.0019, 0.0022]$

Table 10 asks whether the five-feature-zero result is concentrated in one held-out video. The 108 unique windows come from four videos, with 26, 20, 48, and 14 windows, respectively; each window is scored under all four conditions. Every video-by-condition cell has a negative mean on

all four platforms, and all five paired seed differences are negative on each platform. The stronger mixed-condition difference for Unit 2 is a descriptive within-source observation, not an independently replicated effect.

Table 10: Five-feature-zero minus aligned-audio test raster-reconstruction RMSE within each held-out source video. Unit numbers are labels for the four videos, not new sampling units; n counts unique windows per video, each reused across four conditions. Cells give the minimum–maximum of four five-seed platform means, not population intervals. Negative values favor five-feature-zero.

Unit	n	Clean ΔRMSE ↓	Environment ΔRMSE ↓	Music ΔRMSE ↓	Mixed ΔRMSE ↓
1	26	[−0.0124, −0.0123]	[−0.0175, −0.0172]	[−0.0189, −0.0188]	[−0.0173, −0.0169]
2	20	[−0.0135, −0.0127]	[−0.0267, −0.0258]	[−0.0243, −0.0237]	[−0.0334, −0.0326]
3	48	[−0.0161, −0.0158]	[−0.0232, −0.0231]	[−0.0225, −0.0222]	[−0.0204, −0.0199]
4	14	[−0.0134, −0.0132]	[−0.0192, −0.0191]	[−0.0199, −0.0195]	[−0.0185, −0.0177]

Neither table estimates uncertainty over new videos. The video-level comparison covers the complete four-video held-out subset; repeated windows, conditions, seeds, and platforms do not increase its independent-video count. The result warns that output sensitivity to audio or the condition vector does not establish a benefit on this compact visual target.

B RAVDESS raster confirmation at the registered target

The registered RAVDESS speech-video protocol (Livingstone & Russo, 2018) uses an identity raster at 0.25 seconds, mono 16 kHz audio from 1.0 to 1.5 seconds, and a 64×64 target raster at 2.0 seconds. Sixteen actors are assigned to development, four to validation, and four to confirmation. The compact renderer uses a visual encoder, an audio multilayer perceptron, audio-derived scale and bias, and a visual decoder; inputs exclude emotion, intensity, statement, and actor identifiers. The 240 fixed-pair test compares aligned audio with zero audio, audio from another confirmation actor, and a swapped identity raster. Partner actors cycle through the confirmation actors; a deterministic mapping from protocol and clip identity selects the partner clip. Statement and emotion are not matched, so the audio replacement changes actor and utterance together.

The registered protocol and checkpoint score the 2.0-second target on the 240 fixed pairs. Aligned MAE is 0.08990. Control-minus-aligned contrasts are +0.20931 for zero audio, −0.00045 for actor-audio replacement, and +0.04654 for identity-raster replacement. Their paired 95% bootstrap intervals are $[0.20469, 0.21358]$, $[-0.00129, 0.00039]$, and $[0.04362, 0.04926]$. The registered rule requires a strictly positive lower bound for every control; it therefore fails. No fitting or threshold selection used confirmation actors. The actor-audio swap also changes utterance, so it does not identify a speaker effect.

After the registered analysis, we ran an exploratory six-condition analysis on the same 240 unique targets and checkpoint. The cross-actor control uses the registered partner actor but matches modality, vocal channel, emotion, intensity, statement, and repetition in the RAVDESS filename. Another control keeps the actor and changes only the statement code. Neither code match guarantees synchronized phonemes in the 1.0–1.5-second audio window. The new analysis retains each target's six MAEs and groups uncertainty both by pair and by the four confirmation actors. The metadata-matched cross-actor delta is −0.00102, with pair-bootstrap interval $[-0.00176, -0.00029]$ and actor-cluster interval $[-0.00491, 0.00141]$. The actor-specific deltas range from −0.00674 to +0.00176; omitting actor 21 changes the pooled delta to +0.00089. The same-actor statement delta is +0.00004, with pair-bootstrap interval $[-0.00043, 0.00052]$ and actor-cluster interval $[-0.00017, 0.00025]$. The new aligned, zero-audio, original actor-swap, and identity-swap means agree with the registered replay within 0.000001 MAE. These post-confirmation controls are exploratory; neither their pair-level interval nor a retrospective threshold changes the registered decision, and four actor clusters do not support population inference.

To check whether the removal response corresponds to a useful fitted audio signal on this raster endpoint, we separately fit the same AudioFiLM architecture with a constant zero audio tensor for every example. We reused the development and validation actor split, 64×64 target, seed 1701,

optimizer, batch size, 40-epoch maximum, patience, and validation-MAE checkpoint selection. The comparator selected epoch 35; the original selected epoch 18. Both checkpoints score the same 240 confirmation targets. The constant-audio renderer obtains MAE 0.08652 versus 0.08990 for the original. Its paired difference is −0.00338 MAE; the pair-bootstrap interval is [−0.00401, −0.00275] and the four-actor bootstrap interval is [−0.00512, −0.00164]. For seed 1701, all four actor-specific mean differences are negative, from −0.00614 to −0.00143. The original comparator was designed only after the confirmation actors had been opened. Three further matched-seed comparisons used a fixed plan on that same opened set. Table 11 reports every seed; the constant-audio fit has lower MAE in all four. The new training runs use seeds 1702–1704, with all fit and selection settings held fixed except the seed and audio input. They check the observed endpoint but cannot replace a prospectively registered benefit test or imply perceptual video quality. One actor-specific mean is positive in seed 1702 and two are positive in seed 1704; the latter constant-audio fit selected the 40-epoch cap. Actor-level and run-to-run uncertainty therefore remain.

We then repeated all four matched seed pairs with an 80-epoch maximum and the same patience-six validation rule. Every constant-audio fit again has lower MAE on the same targets, with differences ranging from −0.00307 to −0.00070. For seed 1704, the constant-audio checkpoint now selects epoch 48, so this repeat's negative difference is not capped at epoch 40. None of the new fits selects the 80-epoch cap. The new training histories differ from the earlier histories before epoch 40, however; differences between runs cannot be assigned solely to the longer budget. Table 11 shows both sets without treating seeds or reruns as new test populations.

To separate continued fitting from differences between fresh runs, a further frozen check retained the best validation checkpoint available by epoch 40 within each new 80-epoch trajectory, then scored that checkpoint and the final selected checkpoint on the same 240 opened targets. Both arms selected the same checkpoint at or before epoch 40 in seeds 1701–1703: aligned-audio epochs 22, 15, and 19, and constant-audio epochs 34, 29, and 31. In seed 1704, aligned audio stayed at epoch 24, while constant audio selected epoch 38 by the old cap and epoch 48 after continuation. Its MAE fell from 0.08598 to 0.08538, an after-minus-before change of −0.00060. The constant-minus-aligned difference moved from −0.00298 to −0.00358 in that same trajectory. Actor-specific constant-audio changes range from −0.00310 to +0.00077; this single retrospective fit does not establish an actor-population effect. The check uses eight newly fitted trajectories and the same opened confirmation clips. It does not change the registered replacement result or convert the earlier independent 40-versus-80 fits into a controlled budget comparison.

Table 11: RAVDESS same-target sensitivity on 240 fixed targets. Maximum epochs is the stopping cap; validation selection can stop earlier. Δ is constant-audio-fit MAE minus aligned-audio-fit MAE, so negative favors the constant-audio fit. The 80-epoch runs are fresh post-confirmation fits, not exact continuations of the 40-epoch trajectories.

Seed	Max. epochs	Aligned MAE ↓	Constant MAE ↓	Δ MAE ↓
1701	40	0.08990	0.08652	−0.00338
1701	80	0.08803	0.08599	−0.00204
1702	40	0.08979	0.08749	−0.00230
1702	80	0.08874	0.08740	−0.00133
1703	40	0.08984	0.08719	−0.00265
1703	80	0.08841	0.08770	−0.00070
1704	40	0.08853	0.08637	−0.00216
1704	80	0.08873	0.08565	−0.00307

The earlier 1.0-second result is excluded because it does not measure the registered 2.0-second endpoint; all reported RAVDESS results use the 2.0-second target and remain limited to held-out actor raster reconstruction.

C Development checks that stopped visible-speaker routes

These checks explain why the paper reports controlled raster and visual-proxy outcomes instead of a positive visible-speaker result. None opened a sealed confirmation set after its development or validation rule failed.

On AMI meeting recordings (AMI Project, 2005), a registered cross-session word-timing positive control failed. A replacement test of participant headset audio against close-up video found four eligible participants but only two visual-timing positives in one opened session; a second session had only two eligible participants. On HDTF clips (Zhang et al., 2021), a fixed audio-shift timing test passed for two of four development clips and neither of two validation clips. These outcomes stopped the respective source routes before renderer fitting or confirmation; they are endpoint-eligibility findings, not negative held-out renderer results.

On opened GRID development clips (Cooke et al., 2006), an AVA-style evaluator favored aligned over shifted input for two clips and reversed for two; an independent SyncNet checkpoint favored aligned input for one of four. A source-local fixed-crop predictor then passed its directional check for four of six validation talkers, below the preregistered five; its talker-bootstrap interval $[-0.01798, 0.06214]$ crossed zero. The reserved confirmation talkers remained unopened. These development checks do not supply a compatible external evaluator for a visible-speaker statement.